MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 09** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 10: Circles** **MM: 30**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. If PA and PB are tangents to a circle with centre O from point P such that $\angle AOB = 100°$, then $\angle APB$ is equal to: [1]

(A) $50°$ (B) $70°$ (C) $80°$ (D) $100°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (C) $80°$
Justification: $\angle APB = 180° - \angle AOB = 180° - 100° = \boxed{80°}$.

Q2. The incircle of $\triangle ABC$ touches sides BC, CA, AB at D, E, F. If $AB = 10$ cm, $BC = 8$ cm, $CA = 12$ cm, then length AF is: [1]

(A) 5 cm (B) 7 cm (C) 8 cm (D) 9 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 7 cm
Justification: Semi-perimeter $s = \frac{10+8+12}{2} = 15$ cm. Tangent length $AF = s - BC = 15 - 8 = \boxed{7 \text{ cm}}$.

Q3. A chord AB of a circle of radius 10 cm subtends a right angle at the centre O. Tangents at A and B intersect at P. The length of PA is: [1]

(A) 5 cm (B) 10 cm (C) $10\sqrt{2}$ cm (D) $5\sqrt{2}$ cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 10 cm
Justification: $\angle AOB = 90°, \angle OAP = \angle OBP = 90°$, and $OA = OB = 10$ cm, so $OAPB$ is a square. Thus $PA = OA = \boxed{10 \text{ cm}}$.

Q4. Two concentric circles are of radii 15 cm and 9 cm. The length of the chord of the larger circle touching the smaller circle is: [1]

(A) 12 cm (B) 18 cm (C) 24 cm (D) 20 cm

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 24 cm

Justification: Half-chord $= \sqrt{15^2 - 9^2} = \sqrt{144} = 12$ cm $\implies$ Chord $= 2 \times 12 = \boxed{\mathbf{24}\text{ cm}}$.

Q5. How many tangents can be drawn to a circle from a point lying strictly inside the circle? [1]

(A) 0 (B) 1 (C) 2 (D) Infinitely many

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 0

Justification: Every line drawn through an interior point intersects the circle at two distinct points (secant), so $\boxed{\mathbf{0}}$ tangents exist.

Q6. **Assertion (A):** Tangents drawn at the end points of any diameter of a circle are parallel.
Reason (R): Radii to the points of contact are perpendicular to the respective tangents, and consecutive interior angles sum to 180°. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: With $\angle OAP = \angle OBR = 90°$ (or $\angle OAQ = \angle OBS = 90°$), co-interior angles along diameter AB sum to 180°, making the tangents parallel. Both are true and (R) explains (A).

SECTION B: Very Short Answer Type

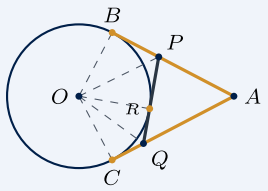
MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. If AB, AC, PQ are tangents to a circle touching at B, C, R respectively and $AB = 11$ cm, find the perimeter of $\triangle APQ$. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Equal Tangent Relations [1 Mark]
From A: $AB = AC = 11$ cm. From P: $PB = PR$. From Q: $QC = QR$.

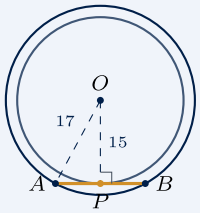
Step 2: Computing Perimeter of $\triangle APQ$ [1 Mark]
Perimeter$(\triangle APQ) = AP + PQ + QA = AP + (PR + RQ) + QA = (AP + PB) + (QC + QA) = AB + AC = 11 + 11 = \boxed{\mathbf{22}\text{ cm}}$.



Q8. In two concentric circles of radii 17 cm and 15 cm, find the length of the chord of the larger circle which is tangent to the smaller circle. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let O be common centre, AB chord touching inner circle at P ($OP \perp AB$, $AP = PB$).



Step 1: Half-Chord via Pythagoras [1 Mark]

$AP = \sqrt{OA^2 - OP^2} = \sqrt{17^2 - 15^2} = \sqrt{64} = 8$ cm.

Step 2: Full Chord Length [1 Mark]

$$\boxed{AB = 2 \times AP = \mathbf{16}\text{ cm}}$$

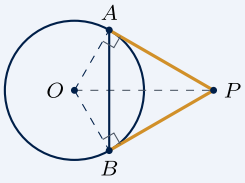
SECTION C: Short Answer Type

MATHS MANTHAN | **(2 × 3 = 6 Marks)**

Q9. In the given figure, OP is equal to the diameter of the circle with centre O. Tangents PA and PB are drawn. Prove that $\triangle PAB$ is an equilateral triangle. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let radius $OA = OB = r$, so $OP = 2r$.



Step 1: Finding $\angle APO$ and $\angle APB$ [1.5 Marks]

In right $\triangle OAP$ ($\angle OAP = 90°$): $\sin(\angle APO) = \frac{OA}{OP} = \frac{r}{2r} = \frac{1}{2} \implies \angle APO = 30°$.

Similarly, $\angle BPO = 30° \implies \angle APB = 30° + 30° = 60°$.

Step 2: Equilateral Property [1.5 Marks]

Since $PA = PB$ (Theorem 10.2), $\angle PAB = \angle PBA = \frac{180° - 60°}{2} = 60°$.

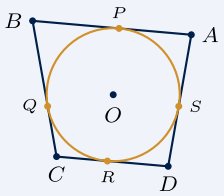
$$\boxed{\therefore \text{All angles are } 60° \implies \triangle PAB \text{ is equilateral [Proved]}}$$

Q10. A quadrilateral $ABCD$ circumscribes a circle. Prove that $AB + CD = BC + DA$. If $AB = 6$ cm, $BC = 9$ cm, $CD = 8$ cm, find the perimeter of the quadrilateral. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Proof of $AB + CD = BC + DA$ [1.5 Marks]

Let circle touch AB, BC, CD, DA at P, Q, R, S. Since $AP = AS, BP = BQ, CR = CQ, DR = DS$: $(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ) \implies AB + CD = BC + DA$. [Proved]



Step 2: Numerical Perimeter [1.5 Marks]

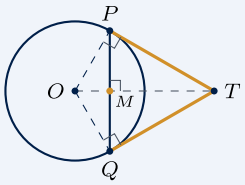
$6 + 8 = 9 + DA \implies DA = 5$ cm.

$$\boxed{\text{Perimeter} = 6 + 9 + 8 + 5 = \mathbf{28}\text{ cm}}$$

Q11. Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that: (i) $\angle PTQ = 2\angle OPQ$, (ii) OT is the perpendicular bisector of chord PQ, (iii) If $\angle PTQ = 60°$ and radius $OP = 6$ cm, find the length of OT. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let OT intersect chord PQ at M.



(i) Prove $\angle PTQ = 2\angle OPQ$: [2 Marks]
Let $\angle PTQ = \theta$. Since $TP = TQ$, $\angle TPQ = 90° - \frac{\theta}{2}$. With $\angle OPT = 90°$, $\angle OPQ = 90° - (90° - \frac{\theta}{2}) = \frac{\theta}{2} \implies \angle PTQ = 2\angle OPQ$. [Proved]

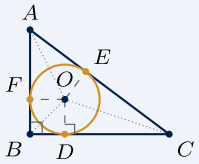
(ii) OT is Perpendicular Bisector of PQ: [1.5 Marks]
In $\triangle TPM$ and $\triangle TQM$: $TP = TQ$, $TM = TM$, $\angle PTM = \angle QTM$ ($\because \triangle OPT \cong \triangle OQT$). By SAS, $\triangle TPM \cong \triangle TQM \implies PM = QM$ and $\angle TMP = \angle TMQ = 90°$. [Proved]

(iii) Length of OT (given $\angle PTQ = 60°, OP = 6$ cm): [1.5 Marks]
Since $\angle PTO = \frac{1}{2}(60°) = 30°$, in right $\triangle OPT$: $\sin 30° = \frac{OP}{OT} \implies \frac{1}{2} = \frac{6}{OT} \implies \boxed{OT = \mathbf{12}\text{ cm}}$.

Q12. A right triangle ABC with sides $AB = 6$ cm, $BC = 8$ cm, and $\angle B = 90°$ circumscribes a circle with centre O and radius r. (i) Find the inradius r. (ii) Find the lengths of the tangent segments AF, BD, CE. (iii) Prove that Area$(\triangle ABC) = r \cdot s$. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let circle touch BC at D, CA at E, and AB at F. Join OA, OB, OC, OD, OE, OF.



(i) Finding Inradius r: [1.5 Marks]
$AC = \sqrt{6^2 + 8^2} = 10$ cm. Since $ODBF$ is a square ($BD = BF = r$): $r = \frac{AB + BC - AC}{2} = \frac{6 + 8 - 10}{2} = \mathbf{2}$ cm.

(ii) Tangent Segments AF, BD, CE: [1.5 Marks]
$BD = r = \mathbf{2}$ cm, $AF = AB - BF = 6 - 2 = \mathbf{4}$ cm, $CE = CD = BC - BD = 8 - 2 = \mathbf{6}$ cm.

(iii) Proving Area$(\triangle ABC) = r \cdot s$: [2 Marks]
Area$(\triangle ABC)$ = Area$(\triangle OBC)$ + Area$(\triangle OCA)$ + Area$(\triangle OAB) = \frac{1}{2}BC \cdot r + \frac{1}{2}CA \cdot r + \frac{1}{2}AB \cdot r = r \cdot \frac{BC + CA + AB}{2} = r \cdot s$. [Proved]

$$\boxed{r = \mathbf{2}\text{ cm}, \quad AF = \mathbf{4}\text{ cm}, \quad BD = \mathbf{2}\text{ cm}, \quad CE = \mathbf{6}\text{ cm}, \quad \text{Area} = r \cdot s}$$

Q13. Ferris Wheel Observation Platform and Tangent Safety Barriers

A giant observation wheel of radius $r = 12$ m is centred at O. From an elevated viewing platform P located 20 m from O ($OP = 20$ m), safety cable barriers PA and PB are strung tangentially to the wheel rim at points A and B.

(i) Calculate the length of the safety cable barrier PA. [1]

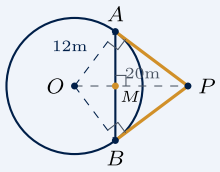
(ii) Find the measure of central angle $\angle AOB$ in terms of $\angle APB$. [1]

(iii) Calculate the area of quadrilateral $PAOB$, OR find the distance from O to the midpoint of chord AB. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) Length of Safety Cable PA: [1 Mark]

In right $\triangle OAP$: $PA = \sqrt{OP^2 - OA^2} = \sqrt{20^2 - 12^2} = \sqrt{256} =$ $\boxed{\mathbf{16}\text{ m}}$.



(ii) Central Angle $\angle AOB$ in Terms of $\angle APB$: [1 Mark]

Since $\angle OAP = \angle OBP = 90°$, $\boxed{\angle\mathbf{AOB} = \mathbf{180°} - \angle\mathbf{APB}}$.

(iii) Main Choice — Area of Quadrilateral $PAOB$: [2 Marks]

Area$(\triangle OAP) = \frac{1}{2} \times OA \times PA = \frac{1}{2} \times 12 \times 16 = 96$ m^2. Since $\triangle OAP \cong \triangle OBP$, $\boxed{\text{Area}(PAOB) = 2 \times 96 = \mathbf{192}\text{ m}^2}$.

— OR — (Alternative Choice — Distance from O to Midpoint M of Chord AB): [2 Marks]

In right $\triangle OAP$, $AM \perp OP$, so $\triangle OMA \sim \triangle OAP$:

$$\frac{OM}{OA} = \frac{OA}{OP} \implies OM = \frac{12^2}{20} = \frac{144}{20} = \boxed{\mathbf{7.2}\text{ m}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★